Queueing Theory :

r.v. $N(t)$: # of customers in queueing system at time t

$N_q(t)$: " " in queue in ——— " t

$N_s(t)$: " " getting service ——— " t.

$$N(t) = N_q(t) + N_s(t)$$

r.v. N : steady state # of customer in queueing system

N_q : " " " " " in queue " " ---

N_s : " " " " " getting service ---

$$N = N_q + N_s$$

$$\Rightarrow \quad E(N) = E(N_q) + E(N_s)$$

$$L = L_q + L_s$$

$L = E(N)$

$L_q = E(N_q)$

$L_s = E(N_s)$

Relationship between time

r.v. w : total time a customer spends in queueing system

$$w = q + s$$

q → time spent in queue

s → service time

$$\Rightarrow \quad E(w) = E(q) + E(s)$$

$$W = W_q + W_s$$

$W = E(w)$

$W_q = E(q)$

$W_s = E(s)$

Cdf

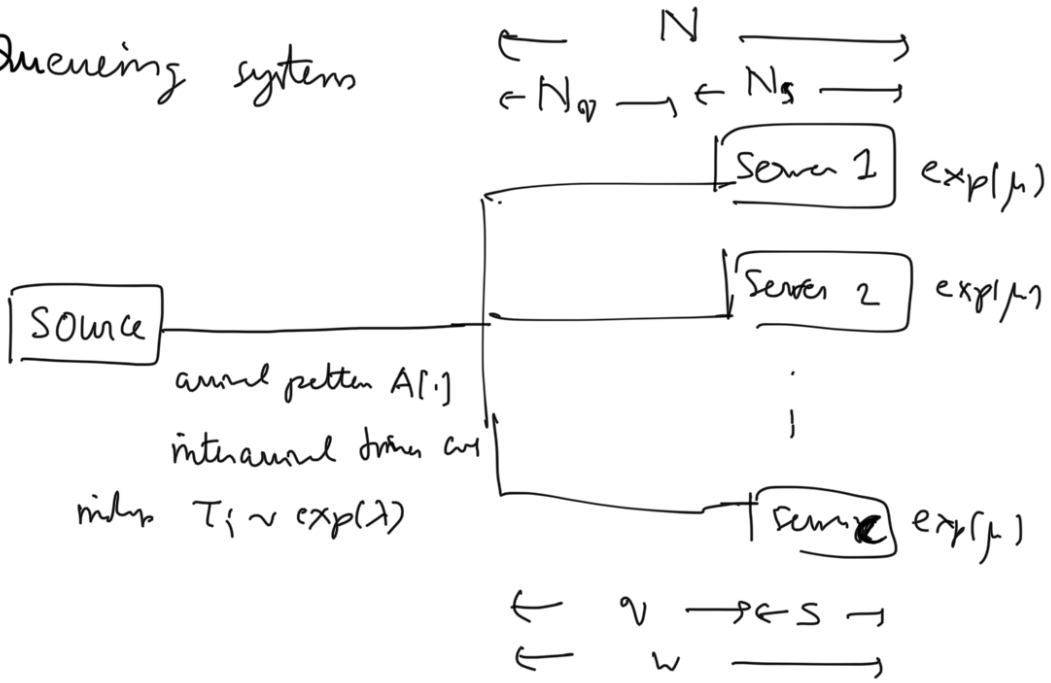
$$W(t) = P(w \le t)$$

$$W_q(t) = P(q \le t)$$

$W_q(t) = P(q \le t)$

$W_s(t) = P(s \le t)$

Queueing systems



If n servers are busy, the T time for next service completion $\sim \exp(n\mu)$

Notation

$A/B/c/K/m/Z$ ← queue discipline FIFO or LIFO

- A: interarrival time distr
- B: service time distr
- c: # of servers
- K: system capacity
- m: # in the popl'n or source

A, B

- GI general indep interarrival time
- G general service time
- M expo. interarrival or service time distr
- U uniform " " " " " "
- D deterministic " " " " " "

$N(t)$ # of customers arrival by time t

av. arrival rate of the customer

$$\lambda_a = \lim_{t \to \infty} \frac{N(t)}{t}$$

<u>Basic cost identity:</u>

$$\begin{bmatrix} \text{av. rate at which} \\ \text{the system earns} \end{bmatrix} = \lambda_a \times \begin{bmatrix} \text{av. amount an entering} \\ \text{customer pays} \end{bmatrix}$$

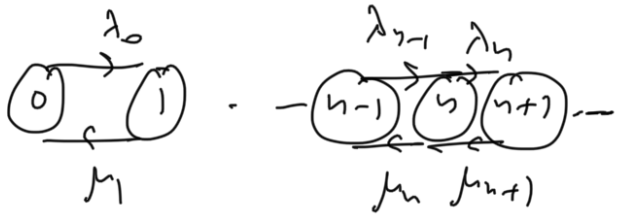
$$L = \lambda_a W$$

$$L_q = \lambda_a W_q \qquad \text{Little's law}$$

$$L_s = \lambda_a W_s$$

— x —

<u>Birth and death process:</u>



In steady state $N = n$

$$P_n = P(N = n)$$

$$S = 1 + C_1 + C_2 + \cdots$$

$$P_0 = \frac{1}{S} \qquad C_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}$$

$$P_n = C_n P_0 \quad , n = 1, 2, \ldots$$

<u>M/M/1 queueing system</u>:

M → interarrival times are $\exp(\lambda)$

M → service times is $\exp(\mu)$

1 → # of server

If $N=n$, then

prob. an of an arrival in time interval of length h is

$$e^{-\lambda h}\lambda h = \lambda h\left(1-\lambda h+\frac{(\lambda h)^2}{2!}----\right)$$

$$= \lambda h + o(h)$$

$$\therefore \lambda_n = \lambda, \quad n = 0,1,2,---$$

$N=n$, service time dist^n

$$W_S(t) = P(S \le t) = 1-e^{-\mu t}$$

If $N=n$,

prob. Service completion in a small interval of length h

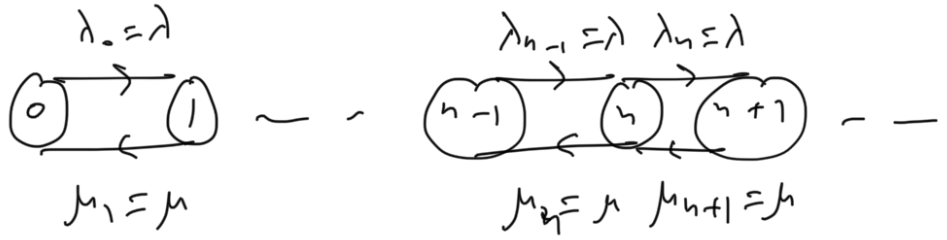
$$= 1-e^{-\mu h} = 1-\left(1-\mu h+\frac{(\mu h)^2}{2!}---\right)$$

$$= \mu h + o(h)$$

B & D process

$$\therefore \mu_n = \mu, \quad n = 1,2,---$$

M/M/1



let $\frac{\lambda}{\mu}=\rho$

$P_0 = \frac{1}{S}$; $S = 1+C_1+C_2+---$

$P_0 = 1-\rho$ $\quad = 1+\rho+\rho^2+--- = \frac{1}{1-\rho}$

$$P_n = P(N=n) = C_n P_0 = \rho^n(1-\rho), \quad n=0,1,2,---$$

$$L = E(N) = \frac{\rho}{1-\rho}$$

$S \sim exp(\mu)$

$$W_S = E(S) = \frac{1}{\mu} = \frac{\rho}{\lambda}$$

$$\sigma_N^2 = \frac{\rho}{(1-\rho)^2}$$

$$\Rightarrow \underline{\rho = \lambda W_S}$$

$$\lambda_a = \lambda \times 1 = \lambda$$

$$L = \lambda W \Rightarrow W = \frac{L}{\lambda} = \frac{\rho}{\lambda} = \underline{W_S} \quad // \quad \frac{1}{\mu(1-\rho)}$$

$$\lambda \qquad \lambda(1-\rho) \qquad 1-\rho$$

$$W_q = W - W_s = \frac{W_s}{1-\rho} - W_s = \frac{\rho}{1-\rho} W_s$$

$$L_q = \lambda W_q = \frac{\lambda \rho}{1-\rho} W_s = \frac{\rho^2}{1-\rho}$$

$$L_s = L - L_q$$

$$P(\text{server is busy}) = 1 - P(\text{server is empty})$$

$$= 1 - P(N=0) = 1-(1-\rho)$$

$$= \rho \rightarrow \text{server utilization}$$

(fraction of time server is busy)

arrival/service ratio $\rho = \frac{\lambda}{\mu}$

<u>Example</u> Suppose that customers arrive at a Poisson rate of one per every 12 min, and that the service time is expo. at a rate of one service per 8 min. Find the parameters of M/M/1 system.

Sol

$$\lambda = \frac{1}{12} \quad \& \quad \mu = \frac{1}{8}$$

$$\rho = \frac{\lambda}{\mu} = \frac{8}{12} = \frac{2}{3} < 1 \qquad \lambda_a = \lambda = \frac{1}{12}$$

$$L = \frac{\rho}{1-\rho} = \frac{2}{3} \times 3 = 2 \qquad W = \frac{L}{\lambda} = 2 \times 12 = 24 \text{ min}$$

$$L_q = \frac{\rho^2}{1-\rho} = \frac{4}{9} \times 3 = \frac{4}{3} \qquad W_q = W - W_s = 24 - 8 = 16 \text{ min}$$

$$W_s = \frac{1}{\mu} = 8 \text{ min}$$

$$L_s = L - L_q = 2 - \frac{4}{3} = \frac{2}{3}$$

$$P_0 = 1 - \rho = \frac{1}{3}$$

$$P_n = P(N=n) = C_n P_0 = \rho^n (1-\rho) = \left(\frac{2}{3}\right)^n \frac{1}{3} ; \; n = 0, 1, 3 \ldots$$

When there is $N=0$ no customer in queueing system then no queueing time seen $W_q(0) = P(q=0) = P(N=0) = 1-\rho$

$N=n$, $q = S_1 + S_2 + \cdots + S_n$, $S_m \sim$ IID $\exp(\mu)$

pdf

$$f_{q|N=n}(t) = \frac{\mu^n}{\Gamma n} e^{-\mu t} t^{n-1}, \quad t > 0$$

For $n > 0$

$$P(q \le t | N=n) = \int_0^t f_{q|N=n}(x)\, dx$$

$$P(0 < q \le t) = E\left(P(0 < q \le t | N)\right)$$

$$= \sum_{n=1}^{\infty} P(0 < q \le t | N=n) P(N=n)$$

$$= \sum_{n=1}^{\infty} \int_0^t \frac{\mu^n}{(n-1)!} e^{-\mu x} x^{n-1} \left(1 - \frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^n dx$$

$$= \int_0^t \lambda e^{-\mu x} \left(1 - \frac{\lambda}{\mu}\right) \underbrace{\sum_{n=1}^{\infty} \frac{(\lambda x)^{n-1}}{(n-1)!}}_{e^{\lambda x}} dx$$

$$= \frac{\lambda}{\mu} \int_0^t (\mu - \lambda) e^{-(\mu-\lambda)x} dx$$

$$= \frac{\lambda}{\mu}\left(1-e^{-(\mu-\lambda)t}\right) = \rho\left[1-e^{-\mu(1-\rho)t}\right]$$

$$= \rho\left(1-e^{-t/W}\right)$$

For $t>0$

$$W_q(t) = P(q=0) + P(0<q\le t)$$

$$= 1-\rho + \rho\left(1-e^{-t/W}\right) = 1-\rho e^{-t/W}$$

Given $N=n$

$$W = S_1 + S_2 + \cdots + S_{n+1} \sim \text{Gamma}(n+1,\mu) \qquad S_r \sim \text{iid } \exp(\mu)$$

pdf

$$f_{W|N=n}(t) = \frac{\mu^{n+1}}{\Gamma(n+1)} e^{-\mu t} t^n, \quad t>0 \qquad \frac{\lambda^\alpha}{\Gamma\alpha} e^{-\lambda x} x^{\alpha-1}$$

$$W(t) = P(W\le t) = E\left(P(W\le t|N)\right)$$

$$= \sum_{n=0}^{\infty} P(W\le t|N=n)\,P(N=n)$$

$$= \sum_{n=0}^{\infty} \int_0^t \frac{\mu^{n+1}}{n!} e^{-\mu x} x^n \left(1-\frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^n dx$$

$$= \int_0^t \mu e^{-\mu x}\left(1-\frac{\lambda}{\mu}\right)\underbrace{\sum_{n=0}^{\infty}\frac{(\lambda x)^n}{n!}}_{e^{\lambda x}} dx$$

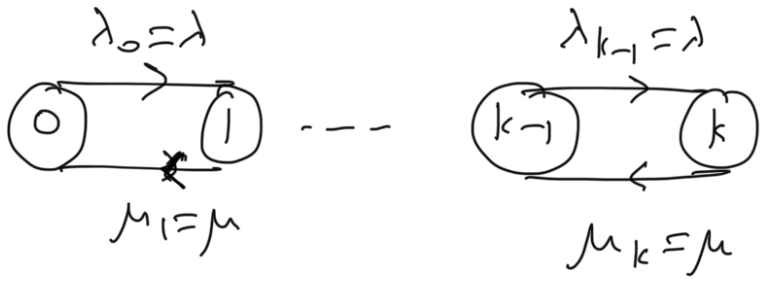
$$= \int_0^t (\mu-\lambda)\, e^{-(\mu-\lambda)x}\, dx$$

$$= 1-e^{-(\mu-\lambda)t} = 1-e^{-\mu(1-\rho)t} = 1-e^{-t/W}$$

$w \sim exp(\frac{1}{W})$ $\quad E(w) = W \; ; \; \sigma_w^2 = W^2$

$s \sim exp(\mu)$ $\quad E(s) = \frac{1}{\mu}$

M/M/1/k queueing system:



B & D process

$$\lambda_n = \begin{cases} \lambda, & n = 0, 1, \ldots, k-1 \\ 0, & n = k, k+1, \ldots \end{cases}$$

$$\mu_n = \begin{cases} \mu, & n = 1, 2, \ldots, k, \\ 0, & n = k+1, k+2, \ldots \end{cases}$$

$a = \frac{\lambda}{\mu}$

✓ $P_n = C_n P_0 = a^n P_0, \quad n = 0, 1, 2, \ldots$

$$\sum_{n=0}^{k} P_n = 1 \Rightarrow P_0 + a P_0 + \cdots + a^k P_0 = 1$$

$$\frac{1 - a^{k+1}}{1 - a} P_0 = 1 \Rightarrow P_0 = \frac{1 - a}{1 - a^{k+1}}$$

if $a \neq 1$

if $a = 1$; $P_0 = \frac{1}{k+1}$

$$P(N=n) = P_n = \begin{cases} \frac{(1-a)a^n}{1 - a^{k+1}} & \text{if } a \neq 1 \end{cases}$$

$$\left\{ \frac{1}{k+1} \right. \qquad \text{if } a=1$$

$a \neq 1$

$$L = \sum_{n=0}^{k} n P_n = \frac{(1-a)a}{1-a^{k+1}} \sum_{n=0}^{k} \underbrace{n a^{n-1}}_{\frac{d}{da} a^n}$$

$$= \frac{(1-a)a}{1-a^{k+1}} \frac{d}{da} \underbrace{\left(\sum_{n=0}^{k} a^n \right)}_{\frac{1-a^{k+1}}{1-a}}$$

$$= \frac{(1-a)a}{1-a^{k+1}} \times \left[\frac{-(1-a)(k+1)a^k + (1-a^{k+1})}{(1-a)^2} \right]$$

$$= \frac{a}{1-a} - \frac{(k+1)a^{k+1}}{1-a^{k+1}}$$

if $a=1$

$$L = \sum_{n=0}^{k} n \times \frac{1}{k+1} = \frac{k}{2}$$

$a = \frac{\lambda}{\mu}$

$$L = \begin{cases} \dfrac{a}{1-a} - \dfrac{(k+1)a^{k+1}}{1-a^{k+1}} & \text{if } a \neq 1 \\ \dfrac{k}{2} & \text{if } a = 1 \end{cases}$$

$$L_s = E(N_s) = E\left(E(N_s \mid N)\right)$$

$$= E(N_s \mid N=0)\, P(N=0) + E(N_s \mid N>0)\, P(N>0)$$

$$= 0 \times P_0 + 1 \times (1-P_0) = 1 - P_0$$

$$L_q = L - L_s = L - (1-P_0)$$

$$P_{blocking} = P_k$$

$$\lambda_a = \lambda \times (1 - P_{blocking}) = \lambda \times (1-P_k)$$

$$W = \frac{L}{\lambda_a}, \quad W_q = \frac{W}{\lambda_a} \qquad W_s = \frac{1}{\mu} = \frac{a}{\lambda}$$

$$a = \frac{\lambda}{\mu}$$

true server utilization $= \rho = \lambda_a W_s = \lambda(1-P_k) W_s$

$$= (1-P_k)\, a$$

For $n = 0, 1, \ldots, k-1$

q_n prob. that an arriving customer who enters the system finds n customers already in the system

event A_n : there are n customers in the system

A : an arrival is about to occur

$$P(A \mid A_n) = \lambda h + o(h)$$

$$P(A) = \sum_{n=0}^{k-1} \underbrace{P(A \mid A_n)}_{\lambda h + o(h)} P(A_n)$$

$$= (\lambda h + o(h)) \left(\sum^{k-1} P(A_n) \right)$$

$$\underbrace{\quad n=0 \quad}_{1-P_k}$$

$$= (\lambda h + o(h))(1-P_k)$$

$$P(N_a = n) = q_n = P(A_n | A) = \frac{P(A \cap A_n)}{P(A)}$$

$$= \frac{P(A|A_n)\,P(A_n)}{P(A)}$$

$$= \frac{(\lambda h + o(h))\,P_n}{(\lambda h + o(h))(1-P_k)} = \frac{P_n}{1-P_k}, \quad n = 0, 1, \ldots, k-1$$

N_a r.v. that counts # of customers in an M/M/1/k system just before a customer arrives to enter the system

$$W(t) = P(w \le t) = E\left(P(w \le t | N_a)\right)$$

$$= \sum_{n=0}^{k-1} P(w \le t | N_a = n)\underbrace{P(N_a = n)}_{q_n}$$

$$= \sum_{n=0}^{k-1} q_n \int_0^t \frac{\mu^{n+1}}{\Gamma(n+1)} e^{-\mu x} x^n \, dx$$

$$= 1 - \sum_{n=0}^{k-1} q_n \int_t^\infty \frac{\mu^{n+1}}{\Gamma(n+1)} e^{-\mu x} x^{n+1-1} \, dx$$

$S_n > t \equiv N(t) \le n-1$

$N(t) \sim PP(\mu)$

$S_n \sim Gamma(n, \mu)$

$$= 1 - \sum_{n=0}^{k-1} q_n \underbrace{\sum_{k=0}^{n} \frac{e^{-\mu t} (\mu t)^k}{k!}}_{Q(n;\mu t)}$$

$$= 1 - \sum_{n=0}^{k-1} q_n \, Q(n;\mu t)$$

$$W_q(t) = P(q \le t)$$

$$= W_q(0) + \sum_{n=1}^{k-1} P(q \le t \mid N_a = n)\, q_n$$

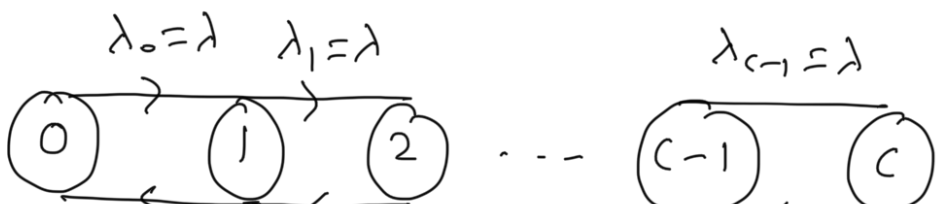
$$= q_0 + \sum_{n=1}^{k-1} q_n \int_0^t \frac{\mu^n}{\Gamma n} e^{-\mu x} x^{n-1} \, dx$$

$$= q_0 + \sum_{n=1}^{k-1} q_n \left[1 - \int_t^\infty \frac{\mu^n}{\Gamma n} e^{-\mu x} x^{n-1} \, dx \right]$$

$$= q_0 + (1 - q_0) - \sum_{n=1}^{k-1} q_n \, Q(n-1;\mu t)$$

$$= 1 - \sum_{n=0}^{k-2} q_{n+1} \, Q(n;\mu t)$$

M/M/c/c queuing system:



$\mu_1 = \mu \quad \mu_2 = 2\mu \qquad\qquad \mu_c = c\mu$

B& D process

$$C_n = \frac{a^n}{n!}, \quad n = 1, 2, \ldots c \quad ; \quad a = \frac{\lambda}{\mu}$$

$$W_s = \frac{1}{\mu}$$

$$S = \frac{1}{P_0} = 1 + C_1 + \cdots + C_c$$

$$= 1 + a + \frac{a^2}{2!} + \cdots + \frac{a^c}{c!}$$

$$P_n = C_n P_0 = \frac{a^n/n!}{1 + a + \frac{a^2}{2!} + \cdots + \frac{a^c}{c!}}, \quad n = 0, 1, \ldots, c$$

$$P_{blocking} = P_c$$

$$\lambda_a = \lambda \times (1 - P_c)$$

$$W_q = 0 \quad ; \quad L_q = 0$$

$$L = E(N) = \sum_{n=\cancel{0}\,1}^{c} n P_n = P_0 \sum_{n=1}^{c} n \frac{a^n}{n!} = a P_0 \sum_{n=1}^{c} \frac{a^{n-1}}{(n-1)!}$$

$$= a P_0 \sum_{n=0}^{c-1} \frac{a^n}{n!}$$

$$W(t) = W_s(t) = 1 - e^{-\mu t} = 1 - e^{-t/W_s}$$